PROOF OF CONCEPT

AMA-KBB

Autonomous Multi-Agent Knowledge Base Builder

LLM-powered research agents + deterministic processing → RAG-ready vector stores

CrewAI · ChromaDB · nomic-embed-text · LiteLLM · Python / uv

THE CHALLENGE

Building domain knowledge bases is manual, slow, and fragmented

Source Discovery

Manually identifying relevant sources across the web takes days of expert effort.

Quality Control

Without domain validation, low-quality or off-topic documents pollute the knowledge base.

RAG Readiness

Raw web content needs cleaning, chunking, and embedding before it's usable in a retrieval pipeline.

THE SOLUTION

A Two-Phase Autonomous Pipeline

PHASE 1 — LLM Research

🔍 Researcher Agent → ResearchPlan

🌐 Scraper Agent → Fetches URLs

✅ Plan Reviewer (SME) validates rubric

🔎 Research Agent → candidate sources

🛡️ Source Reviewer → filters quality

PHASE 2 — Deterministic Processing

🧹 Document Cleaner → normalize text

✂️ Document Chunker → markdown-aware

🧮 Embedding Generator → nomic-embed

🗄️ ChromaDB Storage → vector store

📊 Run Summary → JSON + markdown

ARCHITECTURE

Key Components & Technology Stack

ORCHESTRATION

CrewAI

Multi-agent workflow with YAML-defined agents & tasks

LLM ACCESS

LiteLLM

Multi-provider gateway — swap models via env var

WEB SEARCH

Serper API

Real-time search for candidate source discovery

VECTOR STORE

ChromaDB

Per-run sqlite-backed vector database

EMBEDDINGS

nomic-embed-text-v1.5

State-of-the-art open embedding model

TEXT SPLITTING

LangChain

Semantic, structure-aware chunking

USAGE & OUTPUT

Simple CLI, Rich Artifacts

$ CLI USAGE

```
uv run kbb run \
  --topic "quantum computing"

uv run kbb run \
  --topic "quantum error correction" \
  --rubric rubrics/qec.yaml
```

📁 OUTPUT ARTIFACTS

```
artifacts/run_20260403_151104/

  ├── documents/
  │   └── arxiv.org-abs-2205.md

  ├── chroma.db/
  │   └── chroma.sqlite3

  └── summary.json
```

8	8	8	0
Docs Scraped	Docs Cleaned	Chunks Stored	Filtered Out

Known Limitations & Next Steps

⚠️ Current Limitations

- Paywalled content — academic publishers expose abstracts only
- Search relies on Serper API key (external dependency)
- No PDF fetching — full papers not ingested
- Rubrics must be hand-crafted YAML for each domain

🚀 Next Steps

1. Add PDF fetching via PyMuPDF or pdfplumber
2. Auto-generate domain rubrics with LLM bootstrapping
3. Incremental updates — diff & re-embed changed sources
4. REST API / UI for non-technical knowledge base authoring
5. Evaluation harness — retrieval precision@k metrics

AMA-KBB

From topic to RAG-ready knowledge base — autonomously.

github.com / ama-kbb

CrewAI | LiteLLM | ChromaDB | Serper | nomic-embed | Python uv

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